

Report 3 – Model Accuracy Results

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1. Overall accuracy

Metric	Value
Total samples	3,167
Correct predictions	2,637
Overall accuracy	83.26%

2. Category-wise metrics

For each category: **Support** = number of test samples, **Accuracy** = recall (fraction of that category predicted correctly), **Precision** and **F1** as defined in classification.

#	Category	Support	Correct*	Accuracy	Precision	F1
1	Road Sweeping / Toilet Cleaning / Garbage disposal	607	515	84.84%	86.70%	85.76%
2	Road, pavement, divider, pits, repair / new speed breaker / zebra crossing	551	441	80.04%	89.82%	84.64%
3	Drainage	351	294	83.76%	83.05%	83.40%
4	Water Supply	298	286	95.97%	94.08%	95.02%
5	Street Lights	279	260	93.19%	96.30%	94.72%
6	Encroachments on public premises / roads	219	190	86.76%	84.82%	85.78%
7	Building Permission	141	117	82.98%	76.97%	79.86%
8	Stray Dogs	122	115	94.26%	92.74%	93.50%
9	Fogging / Mosquito nuisance / Dengue malaria disease	107	96	89.72%	90.57%	90.14%
10	Tree Authority	83	75	90.36%	77.32%	83.33%

#	Category	Support	Correct*	Accuracy	Precision	F1
11	Unauthorized hoardigs banners / advertisements on roads / footpath / buildings / directions panels	67	59	88.06%	76.62%	81.94%
12	Garden Cleaning & maintenance	44	26	59.09%	59.09%	59.09%
13	Property Tax Assessment / Payment	38	31	81.58%	86.11%	83.78%
14	Garbage Depot Complaint	36	6	16.67%	28.57%	21.05%
15	Public Health Related	31	19	61.29%	30.65%	40.86%
16	Stray Animals	25	18	72.00%	72.00%	72.00%
17	Bhavan	17	9	52.94%	40.91%	46.15%
18	Environment (H.O)	16	11	68.75%	42.31%	52.38%
19	License (Parwana)	14	1	7.14%	33.33%	11.76%
20	Abhyagat Kaksha	13	5	38.46%	83.33%	52.63%
21	Garden Civil maintenance work	12	8	66.67%	72.73%	69.57%
22	City Development Plan	9	3	33.33%	15.79%	21.43%
23	Traffic Signal	8	6	75.00%	54.55%	63.16%
24	PMC Hospitals treatment	8	7	87.50%	77.78%	82.35%
25	Bridge / Overbridge / Subway / pedestrian	7	5	71.43%	45.45%	55.56%
26	Birth And Death	7	7	100.00%	87.50%	93.33%
27	Property Tax-TAT(Title Transfer)	6	2	33.33%	40.00%	36.36%
28	Traffic-Planning	6	3	50.00%	27.27%	35.29%
29	PMC Properties	6	3	50.00%	37.50%	42.86%
30	Garden Electric maintenance work	6	4	66.67%	100.00%	80.00%
31	Unauthorized banners / advertisements / Permit / license in Pune City	4	2	50.00%	100.00%	66.67%
32	Marriage Registration	4	4	100.00%	100.00%	100.00%
33	General Administration	3	2	66.67%	40.00%	50.00%
34	Primary Education	3	2	66.67%	100.00%	80.00%
35	Information Technology	3	2	66.67%	100.00%	80.00%
36	Sports	3	1	33.33%	33.33%	33.33%

#	Category	Support	Correct*	Accuracy	Precision	F1
37	TDR	2	1	50.00%	100.00%	66.67%
38	Lashkar Water Supply	2	0	0.00%	0.00%	0.00%
39	Communicable Disease	2	0	0.00%	0.00%	0.00%
40	Unauthorized slaughterhouse / crude meat	2	1	50.00%	100.00%	66.67%
41	Urban Poor Scheme	1	0	0.00%	0.00%	0.00%
42	Heritage Cell (Bhavan HO)	1	0	0.00%	0.00%	0.00%
43	RTI Related Complaint	1	0	0.00%	0.00%	0.00%
44	PMC Security Complaints	1	0	0.00%	0.00%	0.00%
45	Vehicle department	1	0	0.00%	0.00%	0.00%
46	Electrical(H.O)	0	0	0.00%	0.00%	0.00%
47	Pension Complaints	0	0	0.00%	0.00%	0.00%
48	Slum	0	0	0.00%	0.00%	0.00%
49	Chawl Complaints	0	0	0.00%	0.00%	0.00%
50	Property Tax-COMPUTER SECTION	0	0	0.00%	0.00%	0.00%

Correct = recall × support (rounded)

3. Summary

Strong categories (support ≥ 5, accuracy ≥ 90%)

Category	Support	Accuracy	F1
Water Supply	298	95.97%	95.02%
Street Lights	279	93.19%	94.72%
Stray Dogs	122	94.26%	93.50%
Tree Authority	83	90.36%	83.33%
Birth And Death	7	100.00%	93.33%

Categories needing improvement (accuracy < 70%, support > 0)

Category	Support	Accuracy	F1
Garden Cleaning & maintenance	44	59.09%	59.09%
Garbage Depot Complaint	36	16.67%	21.05%
Public Health Related	31	61.29%	40.86%
Bhavan	17	52.94%	46.15%
Environment (H.O)	16	68.75%	52.38%
License (Parwana)	14	7.14%	11.76%
Abhyagat Kaksha	13	38.46%	52.63%
Garden Civil maintenance work	12	66.67%	69.57%
City Development Plan	9	33.33%	21.43%
Property Tax-TAT(Title Transfer)	6	33.33%	36.36%
Traffic-Planning	6	50.00%	35.29%
PMC Properties	6	50.00%	42.86%
Garden Electric maintenance work	6	66.67%	80.00%
Unauthorized banners / advertisements / Permit / license in Pune City	4	50.00%	66.67%
General Administration	3	66.67%	50.00%
Primary Education	3	66.67%	80.00%
Information Technology	3	66.67%	80.00%
Sports	3	33.33%	33.33%
TDR	2	50.00%	66.67%
Lashkar Water Supply	2	0.00%	0.00%
Communicable Disease	2	0.00%	0.00%
Unauthorized slaughterhouse / crude meat	2	50.00%	66.67%
Urban Poor Scheme	1	0.00%	0.00%
Heritage Cell (Bhavan HO)	1	0.00%	0.00%
RTI Related Complaint	1	0.00%	0.00%
PMC Security Complaints	1	0.00%	0.00%
Vehicle department	1	0.00%	0.00%

4. Data source

- **Results file:** `report3_model_results.json`
- **Test set:** Report 3 complaints (excluding category "Other")
- **Checkpoint:** `report3_model_results.checkpoint.json` (predictions by `com_id`)